Assignment on File handling

Q1. Write a program that opens a text file and displays the contents of the file.

Q2. Write a program that opens a text file and counts the number of characters, words, and lines in the file.

Q3. Write a program that opens a file and searches for a specific pattern in the file.

Q4. Write a program that opens a file, searches for a pattern, and prints the frequency (number of occurrences) of that pattern in the file.

Q5. Write a program that opens a file, searches for a word, and replaces it with another word throughout the entire file.

Q6. Write a program that opens a file in write mode and adds some text to the file.

Assignment on Database Connectivity.

Q1 Write a program that connects to a **SQLite** database. Create a table named **Employee**, insert sample records into the table, and fetch the details of employees whose salary is greater than 50,000.

Student Information System Project : The **Student Information System (SIS)** is an application designed to manage student-related data efficiently. It integrates multiple modules such as student records, library management, examination, and accounts into a single system.

Student Information System (Using SQLite Database)

1. Project Requirement

Develop a **Student Information System application** that performs basic operations using an **SQLite database** for data storage. The system should be designed using the following classes:

- **Student**
- **Library**
- **Examination**
- **Account**

All data must be stored, retrieved, and managed using **SQLite** (no in-memory storage).

2. Functional Requirements

1. Student Module

- Add new student details to the database
- View student records from the database
- Update student information
- Delete student records

2. Library Module

- Issue books to students (store in database)
- Return books (update database)
- View issued and returned book records

3. Examination Module

- Store student marks in the database
- Calculate and store results/grades
- Retrieve and display exam results

4. Account Module

- Store fee details in the database
- Record fee payments
- Display fee status (paid/balance)

3. Database Requirements (SQLite)

Create an SQLite database with the following tables:

Student Table

- student_id (Primary Key)
- name
- age
- course

Library Table

- book_id
- student_id
- book_name
- issue_date
- return_date

Examination Table

- student_id
- subject
- marks
- grade

Account Table

- student_id
- total_fee
- paid_fee
- balance

4. Operations to Perform

- Insert data into database
- Retrieve data from database
- Update existing records
- Delete records

(All operations must be performed using SQLite queries.)

5. Technical Requirements:

- Use **SQLite** for database management
- Use any programming language (Python/C/Java)
- Implement database connectivity (e.g., `sqlite3` in Python)
- Use SQL commands:
  - `CREATE TABLE`
  - `INSERT`
  - `SELECT`
  - `UPDATE`
  - `DELETE`

6. Expected Outcome : The application should successfully manage student-related data with persistent storage using SQLite and allow users to perform all basic database operations through different modules.